

Machine Learning

Assignment 1 – Polynomial Regression

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GitHub Repo: https://github.com/RyanSajan/ML_Assignment

Introduction

The objective of this assignment is to develop polynomial regression models for two datasets and predict the continuous target variable y . Polynomial regression is used because the relationship between the input features and the target may be nonlinear.

Different polynomial degrees are evaluated using cross-validation to select an appropriate model complexity while avoiding underfitting and overfitting. Mean Squared Error (MSE) is used as the primary model selection metric and R^2 is also calculated to measure the quality of the fit.

Var 1 – Power Plant Steam Turbine Optimization

Approach

The training data is first separated into input features X and target y . Polynomial features are generated using `PolynomialFeatures`. The resulting features are standardized using `StandardScaler`.

For each degree from 1 to 10, a hyperparameter search is performed using 5-fold cross-validation with shuffling. Linear Regression and Ridge regression are evaluated across all degrees. Lasso and ElasticNet were evaluated only for degrees ≤ 6 (var1) and ≤ 9 (var2), since their iterative solvers become computationally expensive and slow to converge on the very large feature sets produced at higher degrees.

The regularization parameters (α and `l1_ratio`) are searched over predefined sets of values. The model configuration with the lowest average cross-validation MSE is selected for each degree, and the overall optimal degree is chosen based on the minimum cross-validation MSE across all tested degrees.

Optimal Degree

The optimal polynomial degree for `var1` is 5. It has chosen 115 out of the 461 features. The following graph shows the cross-validation MSE obtained for the different polynomial degrees.

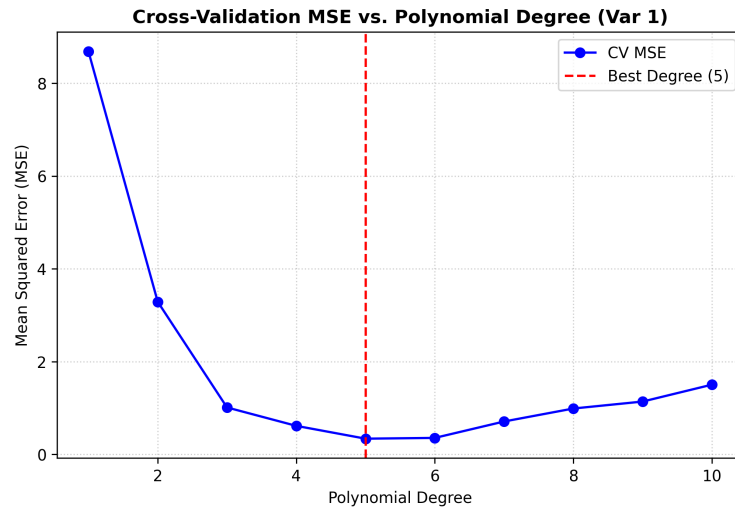


Figure 1: Polynomial degree vs MSE for var1

The degree corresponding to the minimum validation MSE provides the best trade-off between model complexity and prediction error. Increasing the degree beyond this point can introduce unnecessary higher-order and interaction terms and may result in overfitting.

Output

```

=====
Processing Dataset: var1 (1000 rows, 6 features)
=====
Degree 1| Model: lasso      | CV R2 : 0.14116 | CV MSE: 8.6859
Degree 2| Model: lasso      | CV R2 : 0.67494 | CV MSE: 3.2897
Degree 3| Model: elasticnet | CV R2 : 0.89992 | CV MSE: 1.0124
Degree 4| Model: lasso      | CV R2 : 0.93917 | CV MSE: 0.61575
Degree 5| Model: lasso      | CV R2 : 0.96652 | CV MSE: 0.33936
Degree 6| Model: lasso      | CV R2 : 0.96483 | CV MSE: 0.35611
Degree 7| Model: ridge      | CV R2 : 0.92958 | CV MSE: 0.70987
Degree 8| Model: ridge      | CV R2 : 0.90157 | CV MSE: 0.98943
Degree 9| Model: ridge      | CV R2 : 0.88702 | CV MSE: 1.1394
Degree 10| Model: ridge     | CV R2 : 0.85059 | CV MSE: 1.5035

Optimal Configuration Found for var1:
1. Best Degree   : 5
2. Best Model    : LASSO
3. Best Alpha    : 0.01
4. Best L1 Ratio : 1.00
5. Best CV MSE   : 0.33936
6. Best CV R²    : 0.96652

Total features selected: 115 out of 461

```

Figure 2: Output for var1

Var 2 – Subterranean Thermal Reservoir Mapping

Approach

A similar modelling pipeline is used for Var 2, with polynomial features generated for degrees ranging from 1 to 20.

Five-fold shuffled cross-validation with shuffling (`random_state=0`) is utilized across all evaluations to ensure deterministic, reproducible data splits. is utilized for evaluation. Linear Regression and Ridge regression are tested across all 20 degrees, while Lasso and Elastic-Net regressions are additionally incorporated for degrees ≤ 9 to handle the higher risk of overfitting introduced by lower-to-moderate degree expansions.

The configuration yielding the lowest average cross-validation MSE determines both the optimal polynomial degree and the corresponding regularization strength.

Optimal Degree

The optimal polynomial degree for var2 is 10. It has chosen all 285 features. The following graph shows the cross-validation MSE for the different polynomial degrees.

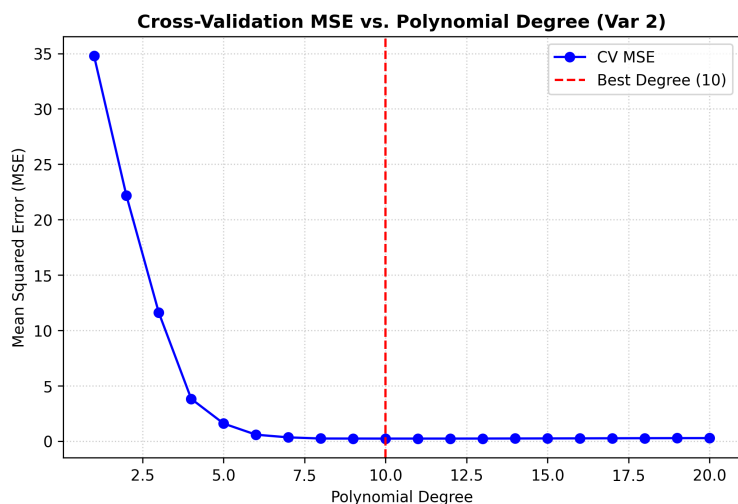


Figure 3: Polynomial degree vs MSE for var2

The minimum of the validation-error curve is used to determine the appropriate model complexity. The use of cross-validation helps prevent choosing a high-degree polynomial simply because it gives a lower training error.

Output

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=====
Processing Dataset: var2 (1000 rows, 3 features)
=====
Degree 1| Model: elasticnet | CV R2 : 0.2644 | CV MSE: 34.8
Degree 2| Model: ridge     | CV R2 : 0.53125 | CV MSE: 22.165
Degree 3| Model: ridge     | CV R2 : 0.7532 | CV MSE: 11.593
Degree 4| Model: elasticnet | CV R2 : 0.91871 | CV MSE: 3.814
Degree 5| Model: elasticnet | CV R2 : 0.96602 | CV MSE: 1.5855
Degree 6| Model: ridge     | CV R2 : 0.98765 | CV MSE: 0.57847
Degree 7| Model: ridge     | CV R2 : 0.99275 | CV MSE: 0.33675
Degree 8| Model: elasticnet | CV R2 : 0.995 | CV MSE: 0.23327
Degree 9| Model: ridge     | CV R2 : 0.99507 | CV MSE: 0.23086
Degree 10| Model: ridge    | CV R2 : 0.99513 | CV MSE: 0.22769
Degree 11| Model: ridge    | CV R2 : 0.99513 | CV MSE: 0.22795
Degree 12| Model: ridge    | CV R2 : 0.99512 | CV MSE: 0.2287
Degree 13| Model: ridge    | CV R2 : 0.99506 | CV MSE: 0.2318
Degree 14| Model: ridge    | CV R2 : 0.99497 | CV MSE: 0.23643
Degree 15| Model: ridge    | CV R2 : 0.99485 | CV MSE: 0.24192
Degree 16| Model: ridge    | CV R2 : 0.99473 | CV MSE: 0.24782
Degree 17| Model: ridge    | CV R2 : 0.99455 | CV MSE: 0.25646
Degree 18| Model: ridge    | CV R2 : 0.9944 | CV MSE: 0.26369
Degree 19| Model: ridge    | CV R2 : 0.99428 | CV MSE: 0.26857
Degree 20| Model: ridge    | CV R2 : 0.99423 | CV MSE: 0.27146

Optimal Configuration Found for var2:
1. Best Degree   : 10
2. Best Model    : RIDGE
3. Best Alpha    : 1
4. Best L1 Ratio : 0.00
5. Best CV MSE   : 0.22769
6. Best CV R2  : 0.99513

Total features selected: 285 out of 285
```

Figure 4: Output for var2

Techniques Used

The main techniques used in the implementation are polynomial feature expansion, feature standardization, Linear Regression, Ridge regression, Lasso regression, ElasticNet regression, and 5-fold shuffled cross-validation.

Polynomial feature expansion allows linear regression models to capture nonlinear relationships. Standardization ensures that all generated features are placed on comparable scales prior to regularization.

Ridge regression adds an L_2 penalty to the model coefficients, while Lasso regression applies an L_1 penalty. ElasticNet combines both L_1 and L_2 penalties, controlled by an l_1 ratio parameter, providing a balanced approach for correlated features. These regularization techniques help control the complexity of polynomial models, particularly when higher-degree expansions produce many correlated terms.

To ensure proper convergence for the iterative coordinate descent solvers used in Lasso and ElasticNet on high-dimensional feature spaces, the maximum number of iterations was set to `max_iter=10000`. Furthermore, Lasso and ElasticNet were restricted to degrees ≤ 6

(for Var 1) and ≤ 9 (for Var 2) to prevent excessive computational bottlenecks during hyperparameter tuning.

The hyperparameter search spaces tested in the implementation are:

$$\alpha_{\text{Ridge}} = \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$$

$$\alpha_{\text{Lasso}} = \{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$$

$$\alpha_{\text{ElasticNet}} = \{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$$

and

$$\text{ll_ratio}_{\text{ElasticNet}} = \{0.1, 0.3, 0.5, 0.7, 0.9\}.$$

Evaluation Metrics

The primary metric used for selecting the model is Mean Squared Error:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

A lower MSE indicates better prediction performance.

The coefficient of determination is also calculated:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

A higher R^2 indicates that the model explains a larger proportion of the variation in the target variable.

Final Model and Conclusion

After selecting the optimal polynomial degree and regularization configuration, the selected pipeline is retrained using the complete training dataset. It is then used to generate predictions for the corresponding test dataset.

The optimal degree is degree 5 for Var 1 and degree 10 for Var 2. The model searches the allowed degree ranges and regularization parameters and selects the configuration with the minimum cross-validation MSE.

Overall, polynomial regression provides a suitable approach for both problems because it can model nonlinear relationships while cross-validation and regularization help control model complexity. The final predictions are saved separately for the two test datasets.